Index calculus on prime-field elliptic curves with smooth maps: a worked tutorial

Petit–Kosters–Messeng (PKC 2016), slides 12–17, and the `pkm-smooth-tower` tool
isogeny-labs/crypto · `research/pkm_smooth_tower_20261114/tutorial/` · 2026-11-14

How to read this. Each idea is followed by a tiny example over $\mathbb{F}_{31}$ or $\mathbb{F}_{101}$ that you can check with pencil and paper. Every example number was produced by `pkm-smooth-tower` and is saved in `tutorial/data/`. The hand checks show where it comes from. This is a teaching document. It makes no claim about how fast the attack is [pcc:unverified], and §9 says what is and is not known.

1. The problem: index calculus needs a factor base

Let $E : y^2 = x^3 + ax + b$ over $\mathbb{F}_p$, P a point of prime order n, and $Q = [k]P$. The **ECDLP** asks for k. Generic algorithms (Pollard rho) take about $\sqrt{n}$ group operations.

Index calculus tries to do better in three steps:

1. Pick a **factor base** $\mathcal{F} \subset E(\mathbb{F}_p)$ of "small" points.
2. Collect **relations**. For random a, b, try to write $R = [a]P + [b]Q = \pm F_1 \pm \cdots \pm F_m$ with every $F_i \in \mathcal{F}$. This is called **decomposing** R.
3. With about $|\mathcal{F}|$ relations, linear algebra modulo n eliminates the F_i and gives k.

Over finite fields $\mathbb{F}_q^*$, "small" means "a product of small primes". Points on an elliptic curve have no such notion. Semaev's idea (2004) was to make "small" mean **the x-coordinate lies in a special set V**:

$$\mathcal{F} = \{(x, y) \in E(\mathbb{F}_p) : x \in V\}.$$

Over an extension field $\mathbb{F}_{q^n}$ one takes $V = \mathbb{F}_q$, the subfield, and Weil descent turns a decomposition into a polynomial system over $\mathbb{F}_q$. A prime field $\mathbb{F}_p$ has no subfield, so it is not clear what V should be.

PKM's answer: let V be the set of roots in $\mathbb{F}_p$ of a **composition of low-degree maps**:

$$V = \{x \in \mathbb{F}_p : L_{n'}(\cdots L_2(L_1(x))\cdots) = c\}, \quad |V| \approx \prod_j \deg L_j.$$

2. Summation polynomials

Semaev's summation polynomials turn "these points add up to O" into an equation in their x-coordinates alone. For three points,

$$S_3(x_1, x_2, x_3) = (x_1 - x_2)^2 x_3^2 - 2((x_1 + x_2)(x_1 x_2 + a) + 2b)x_3 + (x_1 x_2 - a)^2 - 4b(x_1 + x_2).$$

Fact: $S_3(x_1, x_2, x_3) = 0$ exactly when there are signs with $\pm P_1 \pm P_2 \pm P_3 = O$ for some points P_i (over the algebraic closure) with $x(P_i) = x_i$. Read it as "x_3 is the x-coordinate of $P_1 + P_2$ or of $P_1 - P_2$". For fixed x_1, x_2 it is a quadratic in x_3, and its two roots are those two abscissae.

Larger ones are built by resultants: $S_{m+1}(x_1, \ldots, x_m, X) = \mathrm{Res}_U(S_m(x_1, \ldots, x_{m-1}, U), S_3(x_m, U, X))$, of degree 2^{m-1} in each variable. They grow fast. The tool never forms them. It uses the equivalent **chain** with $m - 2$ helper unknowns u_k (the abscissae of the partial sums):

$$S_3(x_1, x_2, u_1) = 0, \quad S_3(u_1, x_3, u_2) = 0, \quad \ldots, \quad S_3(u_{m-2}, x_m, X) = 0.$$

Decomposing R then means: find $x_1, \ldots, x_m \in V$ with $S_{m+1}(x_1, \ldots, x_m, x(R)) = 0$, then lift each x_i to a point and fix the signs.

How big should V be? Each $x \in V$ lifts to a curve point about half the time, and then to two points $\pm$. So $|\mathcal{F}| \approx |V|$. There are about $|\mathcal{F}|^m / m!$ sums of m factor-base points, and the group has about p elements. For a random R to decompose with reasonable probability, take $|V| \approx p^{1/m}$. That is the "$\approx p^{1/m}$" on slides 12 and 14.

3. Why a composition of maps?

You could take V = the roots of one random polynomial of degree $S \approx p^{1/m}$. But then the equation "$x \in V$" has degree S, which is enormous. PKM instead factor the membership test into n' small steps:

$$x_{i,2} = L_1(x_{i,1}), \quad x_{i,3} = L_2(x_{i,2}), \quad \ldots, \quad L_{n'}(x_{i,n'}) = c.$$

Each equation has degree only $q_j = \deg L_j$, at the price of n' new unknowns per point. Their **product** of degrees is the size of V: $|V| = \prod q_j$. For this to work, every step must be **exactly** q_j-to-1 on the values that matter, with all preimages in $\mathbb{F}_p$. That is the point of the two constructions below.

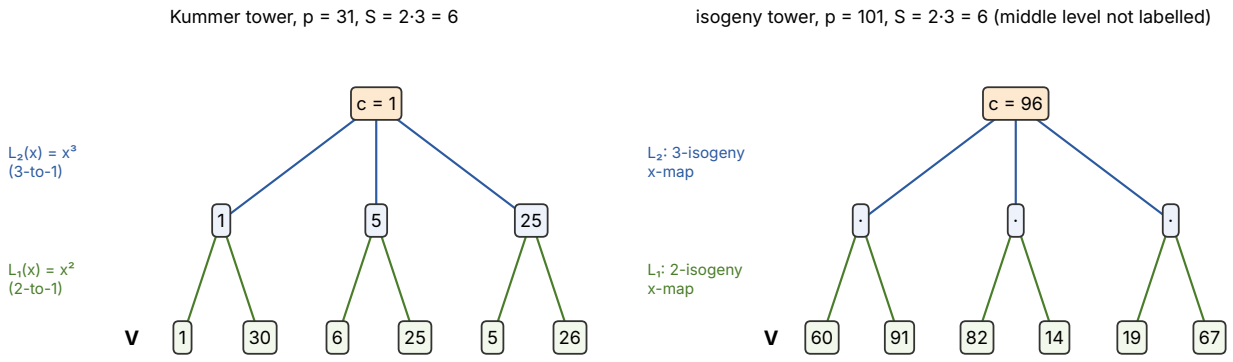


Figure 1: V is the bottom row of a tree of preimages. Each level is one map; a map of degree q has q preimages per node, so $|V| = 2 \cdot 3 = 6$. Left: the Kummer tower of §4. Right: the isogeny tower of §5 (recorded data; its middle level is not labelled).

4. Construction 1: the Kummer tower ("$p-1$ smooth", slide 14)

If $S = q_1 q_2 \cdots q_{n'}$ divides $p-1$, then $\mathbb{F}_p^*$ (cyclic of order $p-1$) contains the group μ_S of S-th roots of unity. Take

$$L_j(x) = x^{q_j}, \quad c = g^S, \quad V = g \cdot \mu_S = \{x : x^S = g^S\}.$$

Each step $x \mapsto x^q$ is exactly q-to-1 on the relevant coset, because $\mu_q \subset \mu_S$.

Example (`data/kummer_p31.json`). $p = 31$, $p-1 = 30$, degrees $(2, 3)$, $S = 6$, $g = 1$. The tool finds $\zeta = 6$ and $V = \{1, 6, 5, 30, 25, 26\} = \mu_6$.

Check by hand. $6^2 = 36 \equiv 5$ and $6^3 = 6 \cdot 5 = 30 \equiv -1$, so 6 has order exactly 6. The first map squares: $1^2 = 30^2 = 1$, $6^2 = 25^2 = 5$, $5^2 = 26^2 = 25$. That gives the middle row $\{1, 5, 25\}$ of the tree, each value hit twice. The second map cubes: $5^3 = 125 = 4 \cdot 31 + 1 \equiv 1$ and $25^3 = 5^6 \equiv 1$. Everything lands on $c = 1$.

Limitation. This needs a large smooth factor of $p-1$. For a given prime that factor might not exist, which leads to the next construction.

5. Construction 2: the isogeny tower (slide 15)

Replace the group $\mathbb{F}_p^*$ (order $p-1$) by the group of an **auxiliary curve** $E'/\mathbb{F}_p$ (order $p+1-t$ for some $|t| \le 2\sqrt{p}$). There are many such curves, so many group orders to choose from.

5.1. The idea

Suppose $E'(\mathbb{F}_p)$ has a cyclic subgroup $G = \langle Q \rangle$ of order S. There is an isogeny $\varphi : E' \to E''$ with kernel exactly G. Pick a point R' with $2R' \notin G$ and set

$$V = x(R' + G) = \{x(R' + [k]Q) : k = 0, \ldots, S-1\}, \quad c = x(\varphi(R')).$$

Why this is the zero set of $x(\varphi(\cdot)) = c$. For $g \in G$, $\varphi(R' + g) = \varphi(R') + \varphi(g) = \varphi(R')$, so every point of $R' + G$ maps to abscissa c. Conversely, if $x(\varphi(P)) = c$ then $\varphi(P) = \pm\varphi(R')$, so $P \in R' + G$ or $P \in -R' + G$.

Both give the same abscissae, since $x(-P) = x(P)$. **Why $|V| = S$.** Two points of $R' + G$ share an abscissa only if one is minus the other: $R' + g = -(R' + g')$, i.e. $2R' \in G$, which we excluded.

5.2. Splitting φ into small steps

Because G is cyclic of order $S = q_1 \cdots q_{n'}$, φ factors into isogenies of prime degree: $\varphi = \varphi_{n'} \circ \cdots \circ \varphi_1$. Step j has kernel generated by $\left[S_j/q_j\right]Q_j$, where $Q_1 = Q$, $Q_{j+1} = \varphi_j(Q_j)$ and $S_j = q_j \cdots q_{n'}$. Each L_j is the x-part of φ_j, a rational function N_j/D_j of degree q_j, given by **Vélu's formula**:

$$L(x) = x + \sum_T \left(\frac{v_T}{x - x_T} + \frac{u_T}{(x - x_T)^2} \right),$$

summing over one point T from each pair $\pm T$ of the nonzero kernel points, with $g_T = 3x_T^2 + a$; $v_T = 2g_T$ and $u_T = 4y_T^2$, except for a 2-torsion point ($y_T = 0$), where $v_T = g_T$ and $u_T = 0$. The image curve is $y^2 = x^3 + (a - 5v)x + (b - 7w)$ with $v = \sum v_T$ and $w = \sum (u_T + x_T v_T)$.

5.3. Worked example at $p = 101$ (`data/isogeny_p101_x50.json`)

Method 1 (§6) found, after 5 random curves,

$$E' : y^2 = x^3 + 66x + 89, \quad \#E'(\mathbb{F}_{101}) = 90 = 2 \cdot 3^2 \cdot 5,$$

with $Q = (76, 40)$ of order $S = 6$ and degrees $(2, 3)$. The tool reports

$$L_1(x) = \frac{x^2 + 63x + 55}{x + 63}, \quad L_2(x) = \frac{x^3 + 79x^2 + 25x + 24}{x^2 + 79x + 20},$$

$R' = (60, 77)$, $V = \{60, 82, 19, 91, 14, 67\}$ and $c = 96$. The middle curve is $E_2' : y^2 = x^3 + 94x + 3$ and the last is $E_3' : y^2 = x^3 + 69x + 75$.

Check by hand. Step 1 (degree 2). The denominator $x + 63$ says the kernel point has $x_T = -63 \equiv 38$. Then $g_T = 3 \cdot 38^2 + 66 = 3 \cdot 30 + 66 = 156 \equiv 55$, so $L_1 = x + 55/(x - 38) = (x^2 + 63x + 55)/(x + 63)$. The codomain has $a = 66 - 5 \cdot 55 = -209 \equiv 94$, and $w = 38 \cdot 55 \equiv 70$ gives $b = 89 - 7 \cdot 70 = -401 \equiv 3$. ✓

Step 2 (degree 3). The denominator is $(x - 11)^2 = x^2 - 22x + 121 \equiv x^2 + 79x + 20$, so $x_T = 11$ on E_2'. Here $y_T^2 = 11^3 + 94 \cdot 11 + 3 \equiv 45$, so $v_T = 2(3 \cdot 121 + 94) \equiv 5$ and $u_T = 4 \cdot 45 \equiv 79$. Then $x(x - 11)^2 + 5(x - 11) + 79 = x^3 + 79x^2 + 25x + 24$. ✓ The codomain has $a = 94 - 25 = 69$ and $b = 3 - 7(79 + 11 \cdot 5) \equiv 75$. ✓

Note how V pairs up under L_1: $R' + [k]Q$ and $R' + [k + 3]Q$ differ by the order-2 kernel point $[3]Q$. So V_0, V_3 share an image, and so do V_1, V_4 and V_2, V_5 (figure above, right).

6. Finding the auxiliary curve (slide 16)

The isogeny tower needs E' with a large **smooth** cyclic subgroup.

Method 1: random curves. Draw random (a', b'), count points, factor the order, and keep the curve if its B-smooth part reaches the target size. The tool records how many curves it tried. At $p = 1000003$, $B = 13$, $S \geq 500$, twenty seeds needed 17.4 curves on average (min 1, max 68; `../data/method1_p1000003_seeds1-20.json`).

Method 2: complex multiplication (CM). Some curves have their order known in advance. Write

$$4p = t^2 + |D| \; v^2$$

with $D < 0$ a discriminant. Cornacchia's algorithm finds (t, v), or proves there is none, in about $\log p$ steps. If there is a solution, the curves with CM by the order of discriminant D have trace $\pm t$, so order $p + 1 \mp t$. For $D = -4$ there are also the traces $\pm 2v$, and for $D = -3$ the traces $\pm(t \pm 3v)/2$ (extra twists). Their j-invariants are the roots mod p of the **Hilbert class polynomial** H_D, which has degree $h(D)$, the class number. For the 13 discriminants with $h(D) = 1$, $H_D = X - j$ with j a known integer: $j(-4) = 1728$, $j(-16) = 287496$, $j(-19) = -884736$, $j(-43) = -884736000$, ...

Example (`data/cm_p101.json`). At $p = 101$ the tool lists 10 CM curves:

D	$4 \cdot 101 = t^2 + \lvert D\rvert v^2$	traces	orders	smooth parts	one curve
-4	$20^2 + 4 \cdot 1^2$	$\pm 20, \pm 2$	82, 122, 100, 104	$2; 2; 2^2 5^2; 2^3$	$y^2 = x^3 + 36x$ (order 100)
-16	$2^2 + 16 \cdot 5^2$	± 2	100, 104	$2^2 5^2; 2^3$	$y^2 = x^3 + 3x + 98$
-19	$10^2 + 19 \cdot 4^2$	± 10	92, 112	$2^2; 2^4 7$	$y^2 = x^3 + 67x + 29$
-43	$19^2 + 43 \cdot 1^2$	± 19	83, 121	none (83 prime, $121 = 11^2$)	$-$

Check by hand. Verify each second column: $400 + 4 = 404$, $4 + 400$, $100 + 304$, $361 + 43$. ✓ Why no $D = -3, -7, -8, -11$? p must be a square mod $\lvert D\rvert$ (roughly), and $101 \equiv 2 \pmod 3$, $101 \equiv 3 \pmod 7$ (squares mod 7 are $1, 2, 4$), $101 \equiv 5 \pmod 8$ and $101 \equiv 2 \pmod{11}$ all fail. Also $1728 \equiv 11 \pmod{101}$: that is the j the tool prints for $D = -4$.

For one fixed p, class number 1 gives only a handful of orders. Allowing $h(D) > 1$ (`--max-disc`) gives many more: at $p = 1000033$, a scan of $\lvert D\rvert \le 1000$, $h \le 16$ found 782 CM curves and a usable $D = -99$ curve, while class number 1 alone found none (`../REPORT.pdf`, §1.3).

Slide 16 also says Method 2 is "more efficient when you can choose p yourself (kind of trapdoor)": pick p so that a CM order is smooth. The tool deliberately only takes p as given and does not do this.

7. Putting it together: the relation system (slide 17)

For m points and n' steps, the unknowns are $x_{i,j}$ ($i \le m$, $j \le n'$), and the equations are

$$\begin{cases} S_{m+1}(x_{1,1}, \ldots, x_{m,1}, X) = 0 & \\ x_{i,j+1} D_j(x_{i,j}) = N_j(x_{i,j}) & 1 \le i \le m,\ 1 \le j < n' \\ N_{n'}(x_{i,n'}) = c D_{n'}(x_{i,n'}) & 1 \le i \le m. \end{cases}$$

(The slide writes the last line as $0 = L_{n'}(x_{i,n'})$, with c folded into the last map.)

Example (`data/system_p101_m2_x50.ms`, msolve format: variables, then p, then equations). The factor base lives on a **different** curve, $E : y^2 = x^3 + 2x + 3$ over $\mathbb{F}_{101}$. The auxiliary E' only supplies V. The target has $X = x(R) = 50$:

```
x1_1,x1_2,x2_1,x2_2
101
(x1_1-x2_1)^2*50^2-2*((x1_1+x2_1)*(x1_1*x2_1+2)+6)*50+(x1_1*x2_1-2)^2-12*(x1_1+x2_1),
x1_2*(1*x1_1+63)-(1*x1_1^2+63*x1_1+55),
(1*x1_2^3+79*x1_2^2+25*x1_2+24)-96*(1*x1_2^2+79*x1_2+20),
x2_2*(1*x2_1+63)-(1*x2_1^2+63*x2_1+55),
(1*x2_2^3+79*x2_2^2+25*x2_2+24)-96*(1*x2_2^2+79*x2_2+20)
```

Line 1 is $S_3(x_{1,1}, x_{2,1}, 50)$. Lines 2–3 say "$x_{1,1} \in V$" in two steps, and lines 4–5 say the same for $x_{2,1}$. Four unknowns, five equations, degrees at most 3 in each tower variable, instead of one equation of degree 6.

For $m = 2$ the tool also solves this directly. For each $x_1 \in V$ it solves the quadratic $S_3(x_1, \cdot, 50) = 0$, keeps roots in V, lifts both to points and **checks by adding them**. It finds one relation:

$$(60, 36) + (67, 83) = R, \quad x(R) = 50.$$

Check by hand. Both points are on E: $36^2 = 1296 \equiv 84$ and $60^3 + 2 \cdot 60 + 3 \equiv 62 + 123 \equiv 84$; $83^2 \equiv 21$ and $67^3 + 2 \cdot 67 + 3 \equiv 86 + 137 \equiv 21$. The chord slope is $\lambda = (83 - 36)/(67 - 60) = 47 \cdot 7^{-1} = 47 \cdot 29 \equiv$

50 (since $7 \cdot 29 = 203 \equiv 1$), so $x_3 = \lambda^2 - 60 - 67 = 2500 - 127 \equiv 76 - 127 \equiv 50$. ✓ Both abscissae, 60 and 67, are in V. Of the six elements of V, only these two lift to E, so this factor base has 4 points.

8. Using the tool

```
# Kummer form (needs S | p - 1)
pkm-smooth-tower kummer  --p 31 --degrees 2,3 --g 1
# Method 1, then the isogeny tower; optional system for m points
pkm-smooth-tower isogeny --p 101 --bound 5 --min-size 6 --seed 1 \
    --curve 2,3 --target-x 50 --m 2 --system sys.ms
# Method 2: class number 1, or Hilbert polynomials with --max-disc
pkm-smooth-tower cm --p 101 --bound 7
pkm-smooth-tower cm --p 1000033 --bound 13 --min-size 300 --max-disc 1000
```

Every tower is checked before it is printed: $|V| = S$ distinct values, and $L(v) = c$ on all of V. For $p < 2^{22}$ it also checks that $L(x) = c$ has no other root in $\mathbb{F}_p$. Code: `src/cryptanalysis/pkm_smooth_tower.rs` (library, 10 unit tests) and `src/bin/pkm_smooth_tower.rs`.

9. What is known, and what is not

Building is easy. Building V, the maps and the system is fast and fully checkable, as in every example here.

What this repository has measured. These are results from earlier studies, quoted from their reports. They have no pcc claim id yet [pcc:unverified].

- **End to end against rho, degree-2 towers.** Index calculus on PKM's Kummer tower ran in the `ecbench` harness, with every group operation charged. On five curves of 20–28 bits it cost **29–141 times Pollard rho** (with the negation map), and the ratio grows with the curve size (`research/pkm_tower_ecbench_20261013/`). The isogeny tower, with its auxiliary-curve search charged, cost 1.00–1.33 times a plain factor base of the same size and stayed **18–136 times above rho** (`research/pkm_isogeny_tower_ecbench_20261015/`).
- **Why the tower did not help there.** Those runs decompose with tables and root finding over V ($m = 2$), and for that V is just a set. The tower base cost the same as a plain base of equal size, within 1 %. In that study's words: "PKM's structure helps an algebraic solver, not a table."
- **The algebraic (Gröbner/F4) side.** Here the tower's shape does matter. On degree-2 Kummer towers, the F4 solving degree climbs slowly with the tower depth (3, 4, 4, 4, 4, 5, 5, 5, 5, 6 at one 31-bit prime; `research/pkm_matched_prime_20261106/`). Regrouping the levels into degree-4 or degree-8 steps cut F4's work by a **constant** factor in three of four cells (×0.17 to ×0.34). In the fourth (×0.54) the study's preset rule found no material effect. In no cell did it change how the work grows. A random system with the same shape gained the same, so the gain comes from the system's shape, not from the curve (`research/pkm_radix_grading_20261104/`).
- **At real sizes.** For P-192 ($m = 4$), the isogeny route's auxiliary-curve search alone is estimated, not measured, at about 2^{74} group operations, against rho's $2^{95.8}$ (`research/pkm_isogeny_tower_ecbench_20261015/`). The deciding cost, solving the systems at $m \geq 3$, has not been measured at any size near that.

Now measured: odd degrees, the auxiliary curve, and the algebraic solver (`research/pkm_odd_tower_f4_20261114/`, EXP-PKM-ODD-001, protocol frozen before data, [pcc:unverified]):

- **Odd degrees help by a constant factor.** At equal $|V|$ and two points ($m = 2$), towers of degree 3 cost 0.24–0.30 times the degree-2 tower's F4 work, and degree 5 0.12–0.16 times. This replicates at a second prime, and a random-coefficient control shows the same gain, so it comes from the shape of the equations. Every arm still grows like $|V|^{3.1\text{–}3.6}$: the constant moves, the growth does not.
- **The auxiliary curve does not help.** Isogeny (rational-map) towers cost 0.76–1.47 times the Kummer towers with the same degrees, with the auxiliary-curve search not even charged.
- **Solving the system loses to enumerating V.** The slides' relation search (F4 plus a linear stage that reads the solutions off the basis) was implemented. It finds exactly the relations enumeration finds (80 of 80 decided targets), but it costs between $2 \cdot 10^2$ and $1.2 \cdot 10^7$ times more $\mathbb{F}_p$ multiplications than simply trying every factor-

base point. The gap grows with $|V|$, at $m = 2$ and at $m = 3$. Larger $m = 3$ systems did not fit in 15 GiB of memory.

Bottom line. At the sizes measured, this attack is slower than Pollard rho by 18–141 times. Solving its polynomial systems is slower than enumerating the factor base by orders of magnitude. Odd degrees shave a constant, but no tower shape tried here changes how the cost grows. The tutorial makes no speed claim.

10. Exercises

1. At $p = 31$, list V for degrees $(3, 2)$ (cube first, then square). Is it the same set as in §4? Is the middle row of the tree the same?
2. Why must $2R' \notin G$? Find what goes wrong at $p = 101$ if you took $R' = Q$.
3. At $p = 101$, $D = -4$ gives four orders. Which ones could support a tower with $S = 25$? With $S = 8$?
4. Using S_3, show that if $x_1 = x_2$ then the roots in x_3 are $x(2P_1)$ and $x(O)$ ("at infinity"), and explain what that means for the quadratic.
5. For $m = 3$, write the summation part of the system with one helper u_1. How many unknowns does the full system have when $n' = 2$?

Answers. (1) The same set: $V = \mu_6$ either way, since $x^6 = 1$ is the composite condition. The middle row becomes the cubes $\{1, 30\}$ instead of the squares $\{1, 5, 25\}$. (2) If $2R' \in G$, two points of $R' + G$ are negatives of each other, so they share an abscissa and $|V| < S$. With $R' = Q \in G$, $V = x(G)$ contains $x(O)$, which is not a field element, and the pairs $\pm[k]Q$ collapse. (3) $S = 25$ needs the order 100 curve ($2^2 5^2$); $S = 8$ needs order 104 ($2^3 \cdot 13$), and the subgroup must be cyclic. (4) $S_3(x_1, x_1, x_3)$ loses its x_3^2 term, so it has degree 1 in x_3: one finite root, which is $x(2P_1)$. The other root went to infinity because $P_1 - P_1 = O$. (5) $S_3\big(x_{1,1}, x_{2,1}, u_1\big)$ and $S_3\big(u_1, x_{3,1}, X\big)$. There are $3 \cdot 2 + 1 = 7$ unknowns.

References. I. Semaev, "Summation polynomials and the discrete logarithm problem on elliptic curves", IACR ePrint 2004/031. C. Petit, M. Kosters, A. Messeng, "Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields", PKC 2016. J. Vélu, "Isogénies entre courbes elliptiques", C. R. Acad. Sci. Paris 273 (1971). G. Cornacchia (1908) on $x^2 + dy^2 = m$.